# Results

# Results I

Results are interpreted only after the registered analysis has run and the deviation ledger (Section 6) has classified any departures from the plan. Every value below is read from `output/data/demo_analysis.json`, which is regenerated by `scripts/generate_figures.py` from the tested `run_registered_analysis` function; the sensitivity row in the final subsection is read from the frozen registration's `sensitivity_analyses`, the same source `build_review_packet` consumes.

## Confirmatory outcome (H1, `primary_score`) I

On the deterministic demonstration data, the registered two-sided permutation test returns:

# Confirmatory outcome (H1, `primary_score`) I

| Quantity | Value |
|---|---|
| Control mean | -0.178991 |
| Treatment mean | 0.823948 |
| Observed mean difference | 1.002940 |
| Permutations | 2000 |
| Shuffles with absolute difference $\geq$ observed | 0 |
| Two-sided permutation p-value | 0.0005 |
| Preregistered `alpha` | 0.05 |
| Confirmatory decision | significant |

## Confirmatory outcome (H1, `primary_score`) I

The observed difference of `1.003` lies far in the right tail of the permutation null distribution: none of the 2000 label shuffles produced a mean difference as extreme, so the add-one-corrected p-value is `0.0005`, below the preregistered `alpha = 0.05`. H1 is therefore **supported** as a confirmatory result. fig. 1 shows the observed statistic against the null it was tested against.

# Confirmatory outcome (H1, `primary_score`) I

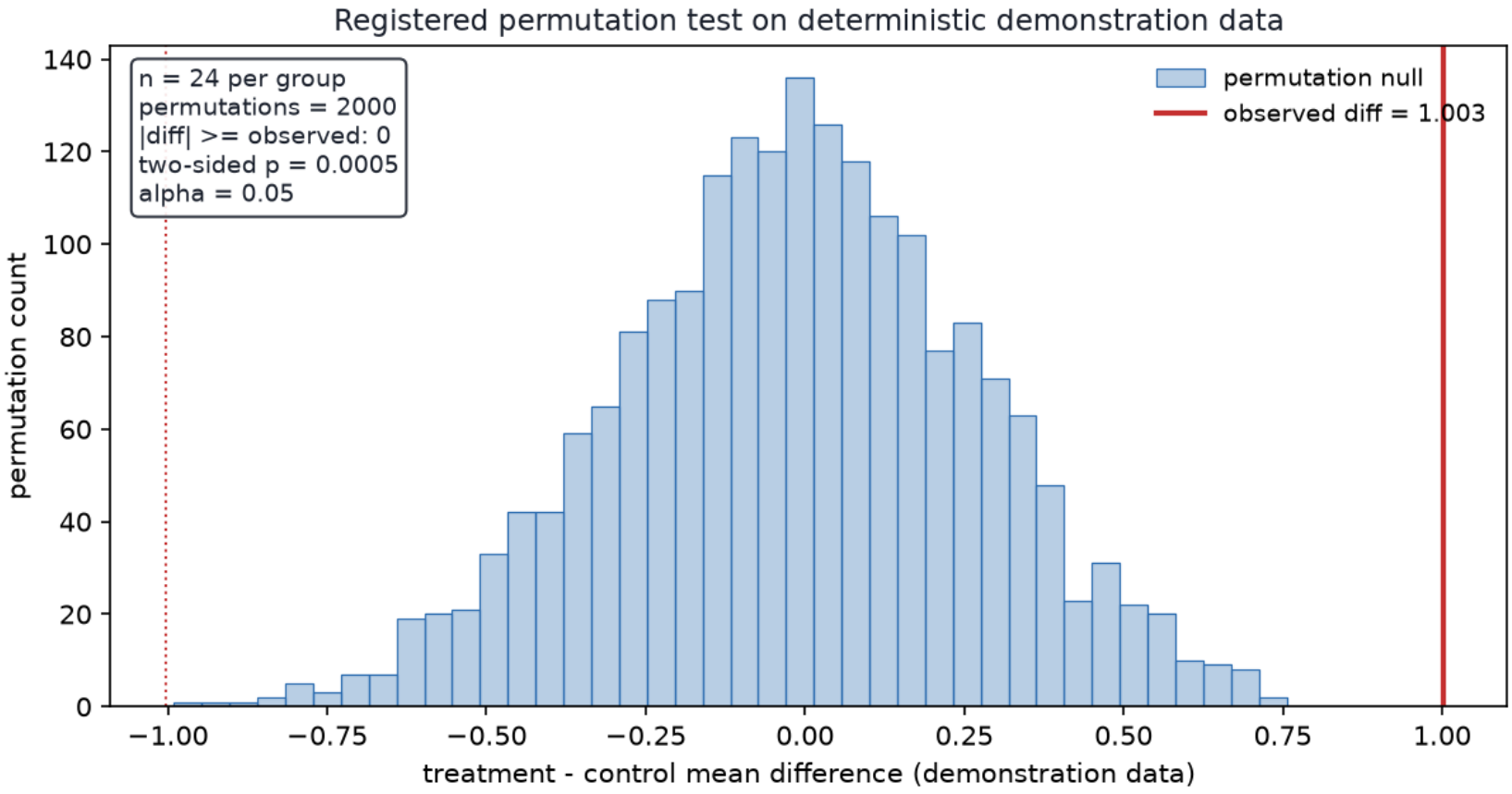



Figure 1: Registered permutation test on the deterministic demonstration data, rendered by `plot_permutation_result` from `permutation_null_distribution` and `run_permutation_test`. The histogram is the label-permutation null distribution of the treatment-minus-control mean difference (2000 seeded shuffles); the solid red line marks the observed difference of `1.003` and the dotted line its mirror image. The annotation reports `n = 24` per group, `0` shuffles at least as extreme, a two-sided p-value of `0.0005`, and `alpha = 0.05`. The observed statistic sits outside the entire null, which is why the primary confirmatory claim holds.

## Claim boundary I

The registration preregisters exactly one confirmatory outcome. The review packet built by `build_review_packet` records `primary_score` as the sole confirmatory outcome and returns an integrity score of `1.0` for a plan-faithful execution. Any other endpoint examined during analysis is exploratory and is reported under Section 6, never here. This separation is enforced in code, not merely asserted in prose.

# Sensitivity analysis I

The frozen registration declares one preregistered sensitivity analysis, and `validate_sensitivity_table` checks every declared row against the registered outcomes and primary model before the review packet is built:

# Sensitivity analysis I

| Analysis | Target | Model | Decision |
|---|---|---|---|
| label_permutation_sensitivity | primary_score | permutation_test | robust |

The row targets the registered primary outcome with the registered primary model and a supported decision, so `validate_sensitivity_table` returns no findings for it and `output/reports/sensitivity_findings.json` records an empty findings list. The same row is carried verbatim into the review packet (`build_review_packet` reads `sensitivity_analyses` from the frozen plan), so the sensitivity check is visible in source, tests, review packets, and this manuscript — never invented after results are known.